$$\frac{x}{(-4)(-5)-(3)(-10)}=\frac{y}{(-10)(4)-(-5)(3)}=\frac{1}{(3)(3)-(4)(-4)}$$

$$\frac{x}{(20)-(-30)}=\frac{y}{(-40)-(-15)}=\frac{1}{(9)-(-16)}$$

$$\frac{x}{20+30}=\frac{y}{-40+15}=\frac{1}{9+16}$$

$$\frac{x}{50}=\frac{y}{-25}=\frac{1}{25}$$

Therefore, we get $x=\frac{50}{25}$; $y=\frac{-25}{25}$

$x=2$; $y=-1$

Thus the solution is $x=2$, $y=-1$.

Verification :

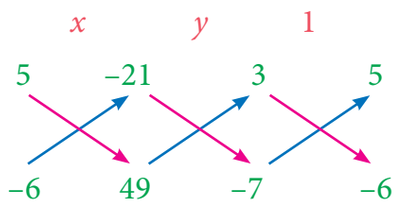
$3x-4y$	=	10	...(1)
$3(2)-4(-1)$	=	10	
$6+4$	=	10	
10	=	10	True
$4x+3y$	=	5	...(2)
$4(2)+3(-1)$	=	5	
$8-3$	=	5	
5	=	5	True

Example 3.53 Solve by cross multiplication method : $3x+5y=21$; $-7x-6y=-49$

Solution

The given system of equations are $3x+5y-21=0$; $-7x-6y+49=0$

Now using the coefficients for cross multiplication, we get,



$$\Rightarrow \frac{x}{(5)(49)-(-6)(-21)}=\frac{y}{(-21)(-7)-(49)(3)}=\frac{1}{(3)(-6)-(-7)(5)}$$

$$\frac{x}{119}=\frac{y}{0}=\frac{1}{17}$$

$$\Rightarrow \quad \frac{x}{119}=\frac{1}{17}\ , \quad \frac{y}{0}=\frac{1}{17}$$

$$\Rightarrow \quad x=\frac{119}{17}\ , \quad y=\frac{0}{17}$$

$$\Rightarrow \quad x=7\ , \quad y=0$$

Verification :

$3x+5y$	=	21	...(1)
$3(7)+5(0)$	=	21	
$21+0$	=	21	
21	=	21	True
$-7x-6y$	=	−49	...(2)
$-7(7)-6(0)$	=	−49	
-49	=	−49	
-49	=	−49	True

Note

Here $\frac{y}{0}=\frac{1}{17}$ is to mean $y=\frac{0}{17}$. Thus, $\frac{y}{0}$ is only a notation and it is not division by zero. It is always true that division by zero is not defined.

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